Cooling and Launching Orbital Data Centers: Concept to Architecture in 12 Days

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An engineer sets the requirements. An AI agent runs the analysis. Physics solvers check the selected design.

Human requirements / design decisions → AI agent (closed source) → Open analysis scripts Apache-2.0 → GPU: Warp, PhysicsNeMo, Newton CPU: CalculiX FEA → Solver checks → Human review / approval

YouTube GitHub

PROBLEM

Orbital data centers can ease power generation concerns on Earth, but their satellites must be mass and volume efficient for launch and meet radiative cooling needs. SpaceX's published targets of 250 kW peak / 175 kW average per satellite and 75 kW per ton, with a 100 t Starship launch, imply 7.5 MW installed peak / 250 kW = 30 satellites.

Why it matters: AI agents and GPU acceleration allow rapid digital prototyping; 12 coworking days for an engineer to go from concept to a chosen mechanical and thermal architecture ready for detailed design.

DESIGN ARCHITECTURE (PER SATELLITE)

Compute: one Vera Rubin NVL72 (72 GPUs + 36 CPUs); 250 kW peak at beginning of life. Assumed 640.38 kg payload allocation in a 600 × 1200 × 937 mm box.

Launch fit: All 30 satellites fit in one Starship: 30 × 3,026 kg = 90.8 t + 2 t dispenser + 7.2 t reserve = 100 t; stowed stack 15.6 m tall in the modeled 17.24 m payload envelope (≥3.9 mm radial margin).

Orbit: 600 km dawn-dusk sun-synchronous. Arrays face the Sun; radiators assumed edge-on to it, radiating from both faces.

Power: 862 m² solar array; assumed module rating 324.55 W/m² on its remaining planar area; 175 kW in sunlight and a 164.4 kW annual average, including eclipse.

Thermal: 140 radiator panels; coating emissivity 0.92. Fins and tubes selected by search; modeled glycol-water loop. Cold-plate resistance assumed for Rubin GPU and HBM.
175 kW in sunlight: rack coolant 339 K to 359 K; Rubin GPU est. 371 K (target ≈370 K), HBM 373 K (limit 378 K).
250 kW peak (steady state): rack coolant 370 K to 390 K; Rubin GPU est. 410 K, HBM est. 412 K.

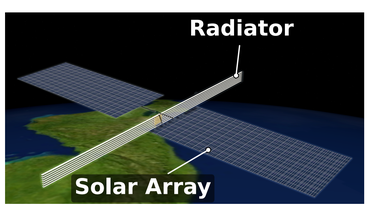
Rack structure: Compute rack screening (CalculiX beam model, 6 g axial / 2 g lateral, FS 1.25): a ≈46 kg aluminum frame inside the 640 kg allocation reaches 35 Hz (rack design target) with large strength margins.

Panel vibration: Stowed radiator-panel first mode 128–153 Hz across support-pad assumptions (≈139 Hz nominal), well above the 35 Hz reference.

Folding and deployment: All 36 representative hinges lock in a rigid Newton simulation; a sampled clearance check finds no collisions.

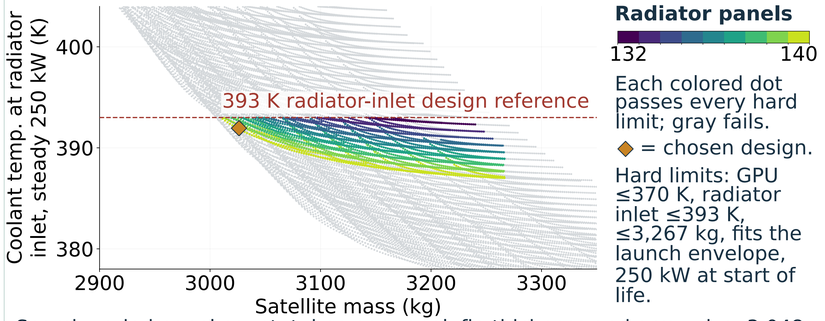


Simulated release: 30 satellites leave the launch vehicle through a side slot and unfold (kinematic, assumed parameters).



Illustrative Omniverse Isaac Sim RTX render, contrast enhanced; not flight validation. Earth: NASA Blue Marble.

1 Radiator search and sensitivity | Warp



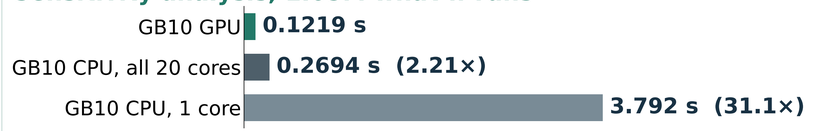
Search varied panel count, tubes per panel, fin thickness and array size. 3,048 of 1.81M designs meet every hard limit; lightest with 1 K extra margin chosen, then 1.05M what-if runs co-varied 14 inputs over assumed ranges, not measured probabilities. At 175 kW in sunlight (screening model): GPU median 368.7 K, under 370 K in 65% of runs; HBM under 378 K in all runs. Radiator emissivity and power conversion efficiency move temperatures most; cold-plate resistance mainly moves the GPU and rack temperature rise mainly moves the coolant.

2 GPU-accelerated design search | GB10

Search, 1.81M designs



Sensitivity analysis, 1.05M what-if runs

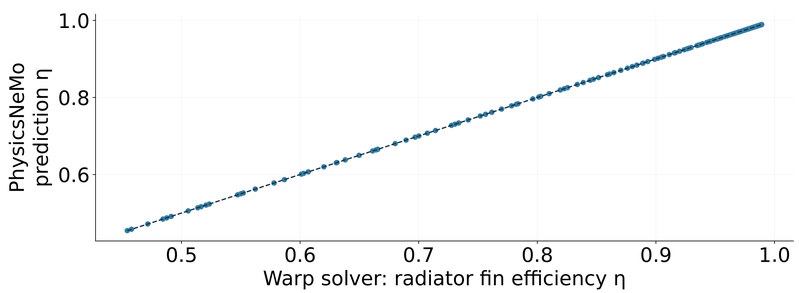


32.9× faster than one GB10 CPU core; 2.35× faster than all 20 cores.

Versus one core, the GPU pulls ahead above about 1,000 designs and levels off at 32–33× from about 260,000 designs to the full 1.81M.

Same double-precision math on all; median of 3 runs; timing excludes setup and compile. GPU: Warp kernels on CUDA. CPU: 1 Arm Cortex-X925 core in the GB10 Grace Blackwell, 1 thread. All cores: the same Warp fp64 code on all 20 cores (10 X925 + 10 A725), not hand-tuned CPU code.

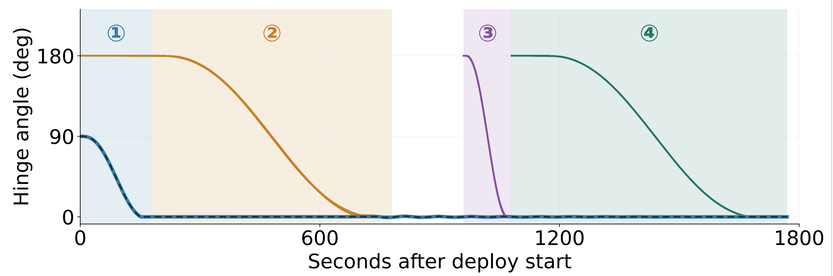
3 AI fin model check | PhysicsNeMo / Warp



A PhysicsNeMo neural model learns radiator fin efficiency from Warp solver results (384 training fins), then is checked on 128 fins it never saw, so a fast learned model can stand in for the solver when screening fin shapes.
Max |Δη| = 9.3e-5.

The chosen design's fins were re-solved with Warp, so no final number rests on the learned model. A Warp autodiff optimizer, treating panel count and fin size as continuous, comes within 0.65% of the search's lightest radiator. Its gradients match finite differences; it is a local check, not a global optimum.

4 Unfolding after release | Newton / Warp



① 0–180 s: solar-array root hinge swings each wing out from the body.
② 180–780 s: solar-array fold-out hinges open the folded panels accordion-style.
③ 960–1,080 s: solar-array width hinges spread the wing to full width.
④ 1,080–1,770 s: radiator hinges unfold the radiator.

Newton simulates the unfold; Warp re-solves all 21 solar-array hinges (root hinge dashed); they agree within 0.0061°. All 36 representative hinges lock; no panel-to-panel collisions in a separate clearance check. Rigid panels, assumed hinge parameters.

Targets: spacex.com/spacexai/starmind. Independent study from public information; not affiliated with or endorsed by SpaceX.
A basis for further detailed design. Simulation study, not flight qualification.